Plumbfinder

EXECUTIVE SUMMARY: PLUMBFINDER

Problem

Water damage inside homes is a large and growing problem in the United States. Alot of it comes from slow hidden leaks that go on for weeks or months inside walls, under floors and under slabs before anyone notices. By the time a homeowner sees a stain on the ceiling or gets an unexpectedly high water bill, the damage is already done and the repair can be many thousands of dollars. Older homes are hit the hardest because they still have original pipes, like galvanized steel and old copper with soldered joints, that corrode and fail over time. There are millions of such homes in the country and the owners are often on fixed budgets and do not have a good way to know whats going on behind the drywall.

Today homeowners have two options and neither is good. They can wait and react when something goes wrong, or they can pay a plumber or leak detection specialist who charges a few hundred dollars per visit and only finds the problem if it is leaking at that moment. Existing smart home leak sensors only tell you there is water on the floor, which is already too late, and they do not tell you where the pipe is failing. Our team has talked to homeowners and many told us they just feel anxious about it and do not know what to do.

Solution

Plumbfinder is a smart attachment for the water meter plus a mobile app. The attachment clips onto the meter or the main supply line in a few minutes with no plumber and no cutting of pipes. Inside it there is a sensor package that reads flow, pressure and vibration of the water many times per second. The data goes by wifi to our cloud, where our analytics software looks for the small patterns that show a leak, for example tiny flow at night when nobody is using water or a specific vibration frequency from a pinhole in a pipe. The software then compares the pattern to a library of leak signatures and gives a best guess of the zone of the house where the failing pipe is, and how serious it is. The homeowner gets a notification on the app that says something like probable slow leak in the bathroom supply line, medium urgency, and can then call a plumber who knows where to look instead of tearing open walls.

We believe this approach is differentiated because it combines a physical product with data analytics and gives location, not only a yes or no alarm. Our algorithms improve as we get more homes in the system, which creates a data advantage over time.

Market

According to public housing data, there are roughly 80 million homes in the US built before 1990, and we think a good share of them have pipe types at risk. We have not yet done a full market sizing but as a rough estimate if even 5 percent of these homes adopted a product like ours at around 150 dollars of hardware and 8 dollars per month, the market would be in the billions. Our initial target is single family owner occupied homes built between 1940 and 1980 in cities with older housing stock and hard water, starting with our own region. Secondary customers are landlords with small rental portfolios, home inspectors, and insurance companies who pay out for water claims.

Business Model

We plan to make money in three ways. First, the hardware is sold direct to consumers at about 149 dollars, with a margin we estimate at 35 to 40 percent at current prototype costs, which should improve with volume. Second, a monthly subscription of 7.99 dollars for the monitoring, app and alerts, which gives recurring revenue. Third, in the future, partnerships with insurance carriers who could subsidize devices or give premium discounts, and referral fees from plumbers who get leads from our alerts. The insurance channel is the most promising but we have only had early conversations and no agreements yet.

Competition

There are several competitors in the smart water space. Some companies sell whole home shutoff valves that cut the water when a leak is detected, and others sell flow monitors that attach to the main line. Basic leak sensor pucks are cheap and widely available. These are all good products but in our understanding most of them focus on detecting that a leak exists or stopping it, and not on telling the homeowner where the pipe is failing inside the house. Some of them also require a plumber to install and cost much more. Our weakness is that we are small, have no brand, and larger companies could add a location feature if they wanted to. Our accuracy for locating leaks is also not yet proven at scale, which is our main technical risk.

Team

Dariusz Kowalczyk is our CEO and is a senior in mechanical engineering, who led the design of the hardware and the enclosure and has summer experience at a local manufacturing firm. Marisol Okonkwo-Reyes is our CTO, a junior in computer science with a focus on machine learning, who built the app and the leak detection software. Teague Brandvold is our head of business and is a senior in business administration, who runs customer interviews, pricing and partnerships. We also have an informal advisor, a retired plumbing contractor with 30 years of experience, who has helped us understand real leak situations and who lets us test in homes he knows.

Milestones and Financials

To date we have built a working prototype of the attachment and a first version of the app, we registered the company as an S Corp, and we have a website. We have placed units in 9 homes owned by friends, family and neighbors of our advisor, and so far the system has flagged 3 leak events, 2 of which were confirmed by a plumber and one was a false alarm. This is encouraging but the sample is very small and we know it is not a real validation. We have spent about 6,500 dollars so far, mostly on parts, funded from our own savings and a small university grant. Our current cost per unit is around 92 dollars in small batches. We expect in year one to sell perhaps 800 units if we get a good channel, which would be about 120,000 dollars of hardware revenue, plus subscription revenue that builds up during the year, but these are early estimates and we do not have a full financial model yet.

What We Are Asking For

We are asking for prize funding and support from this competition. Specifically we would use funds to build a pilot batch of 100 units, improve the accuracy of our location algorithm through testing in more homes with different pipe types, get the electronics tested for safety certification, and hire a part time contractor to help with the cloud backend. Beyond the money we would value mentorship, especially introductions to insurance industry contacts and advice on how to set up our financial model and our pricing. We believe Plumbfinder can help many homeowners avoid costly damage and we are excited to take the next step.